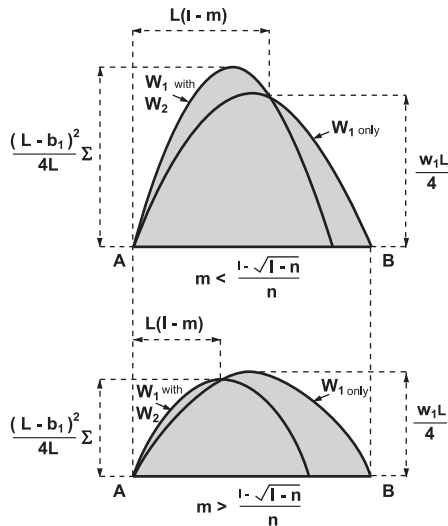


I-BEAMS

Simply supported beam



Two concentrated moving loads (continued)

If $m > n$, the Maximum Bending Moment at any section always occurs under W_1 (the heavier load), whether W_2 is on or off the span.

For a section distance X from A:

$$X \leq L(1 - m) \quad : \quad L(1 - m) \leq X$$

$$M_{\max} = \frac{(L - b_1 - X)X}{L} \quad \Sigma \quad : \quad M_{\max} = \frac{(L - X)X}{L} W_1$$

If $n < m < \frac{L - \sqrt{L - n}}{n}$ the Absolute Maximum Bending

Moment occurs under W_1 with W_2 on the span.

If $n < m > \frac{L - \sqrt{L - n}}{n}$ the Absolute Maximum Bending

Moment occurs under W_1 at mid-span with W_2 off the span.

Note: When the two loads are equal ($W_1 = W_2$ and $n = 1/2$)

the critical value of $\frac{L - \sqrt{L - n}}{n} = 0.5858$.

Simply supported beams carrying several moving concentrated loads

The Maximum Reaction and the Maximum Shear due to several moving concentrated loads occur at one support with one of the loads at that support. The location producing the Absolute Maximum must be found by trial.

The Maximum Bending Moment due to several moving concentrated loads occurs under one of the loads when that load and the gravity centre of all loads are equidistant from mid-span. The Absolute Maximum must be determined by trial.